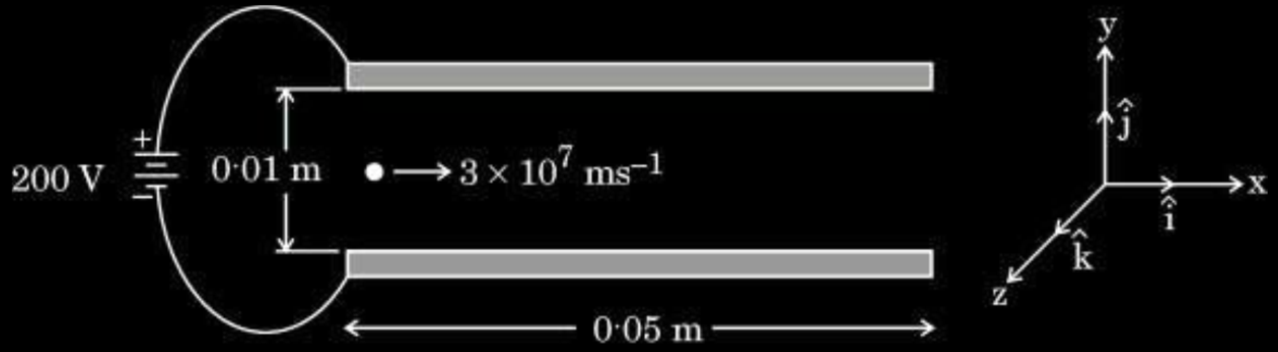
29. The electric potential (V) and electric field (E) are closely related concepts in electrostatics. The electric field is a vector quantity that represents the force per unit charge at a given point in space, whereas electric potential is a scalar quantity that represents the potential energy per unit charge at a given point in space. Electric field and electric potential are related by the equations $E_r = \frac{-dV}{dr}$ and $\vec{E} = E_r \hat{r}$, i.e., electric field is the negative gradient of the electric potential. This means that electric field points in the direction of decreasing potential and its magnitude is the rate of change of potential with distance. The electric field is the force that drives a unit charge to move from higher potential region to lower potential region and electric potential difference between the two points determines the work done in moving a unit charge from one point to the other point. CBSE 2026

A pair of square conducting plates having sides of length 0·05 m are arranged parallel to each other in x-y plane. They are 0·01 m apart along z-axis and are connected to a 200 V power supply as shown in the figure. An electron enters with a speed of 3×10^7 ms^{-1} horizontally and symmetrically in the space between the two plates. Neglect the effect of gravity on the electron.



(i) The electric field $\vec{E}$ in the region between the plates is :

(A) $\left(2 \times 10^2 \dfrac{V}{m}\right)\hat{k}$

(B) $-\left(2 \times 10^2 \dfrac{V}{m}\right)\hat{k}$

(C) $\left(2 \times 10^4 \dfrac{V}{m}\right)\hat{k}$

(D) $-\left(2 \times 10^4 \dfrac{V}{m}\right)\hat{k}$

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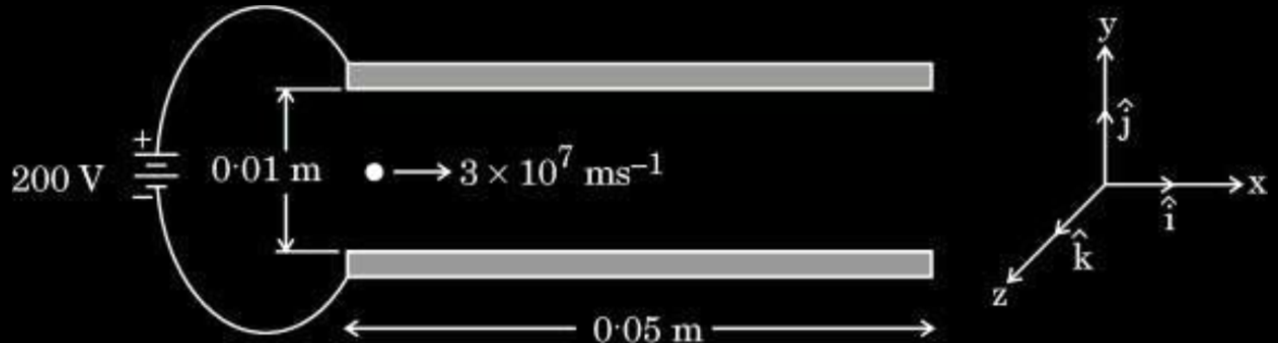

$\boxed{\text{(D) } (-2 \times 10^4 \text{ V/m}) \ k}$

(ii) In the region between the plates, the electron moves with an acceleration $\vec{a}$ given by : *1*

(A) $-(3\cdot5 \times 10^{15}\ ms^{-2})\ \hat{k}$

(B) $(3\cdot5 \times 10^{15}\ ms^{-2})\ \hat{k}$

(C) $(3\cdot5 \times 10^{13}\ ms^{-2})\ \hat{i}$

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(D) $-(3{\cdot}5 \times 10^{13}\ ms^{-2})\ \hat{i}$

(B) $(3.5 \times 10^{15}\ m/s^2)\ k$

(iii) (a) Time interval during which an electron moves through the region between the plates is : *1*

(A) $9 \cdot 0 \times 10^{-9}$ s (B) $1 \cdot 67 \times 10^{-8}$ s

(C) $1 \cdot 67 \times 10^{-9}$ s (D) $2 \cdot 17 \times 10^{-9}$ s

(iii) (a) Time interval during which an electron moves through the region between the plates is : *1*

(A) $9{\cdot}0 \times 10^{-9}$ s (B) $1{\cdot}67 \times 10^{-8}$ s

(C) $1{\cdot}67 \times 10^{-9}$ s (D) $2{\cdot}17 \times 10^{-9}$ s

(C) 1.67×10^{-9} s

(b) The vertical displacement of the electron which travels through the region between the plates is : *1*

(A) 10 mm (B) 4·9 mm

(C) 5·9 mm (D) 3·0 mm

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(A) 10 mm (B) 4·9 mm

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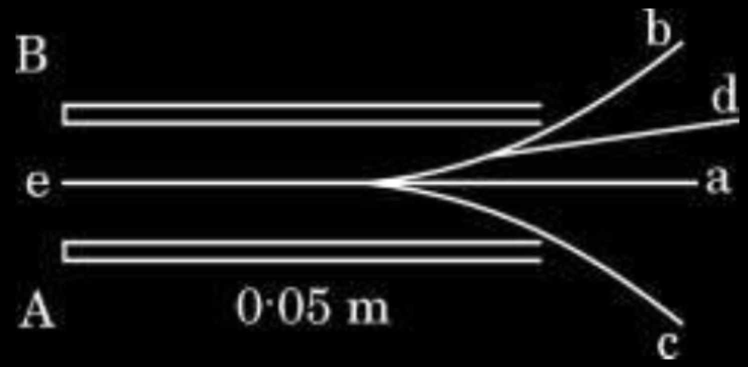
$\boxed{\text{(B) 4.9 nm}}$

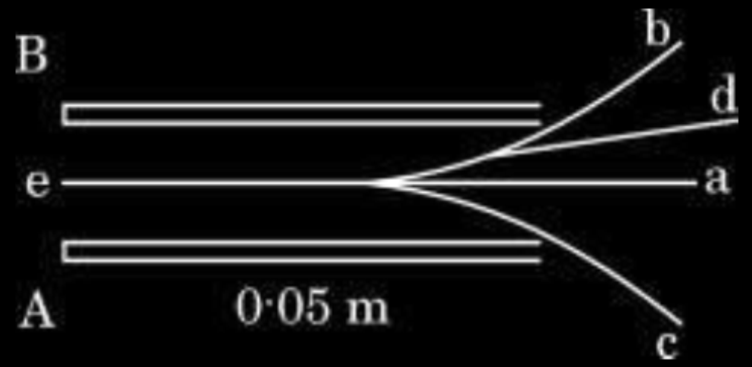
(iv) Which one of the following is the path traced by the electron in between the two plates ? *1*



(A) a (B) b

(C) c (D) d

(iv) Which one of the following is the path traced by the electron in between the two plates ? *1*

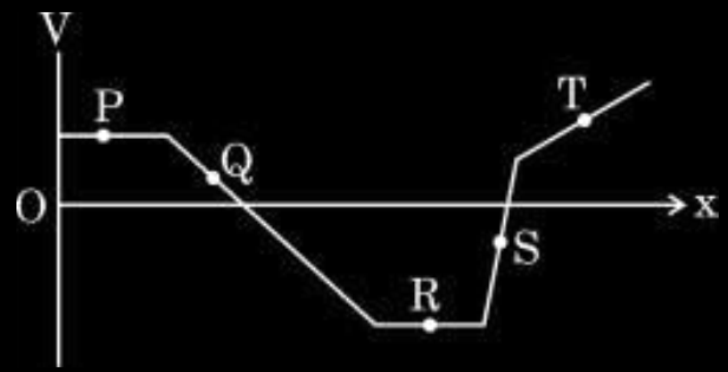


(A) a (B) b

(C) c (D) d

(B) b

2. The figure represents the variation of the electric potential V at a point in a region of space as a function of its position along the x-axis. A charged particle will experience the maximum force at :



(A) P

(B) Q

(C) R

(D) S

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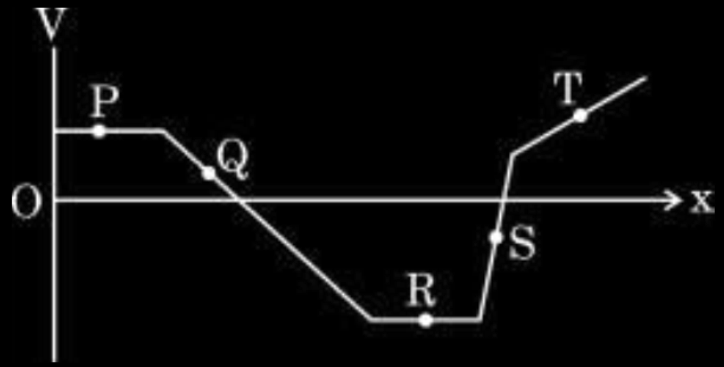

(A) P

(B) Q

(C) R

(D) S

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(D) S

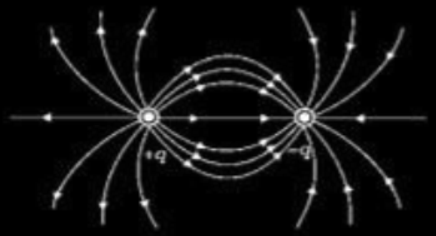
25. (i) Draw electric field lines and equipotential surfaces for a system of two equal and opposite point charges separated by some distance. 3

(ii) Why electric field $\vec{E}$ at a point on an equipotential surface must be perpendicular to the surface at that point ?

25. (i) Draw electric field lines and equipotential surfaces for a system of two equal and opposite point charges separated by some distance. **3**

(ii) Why electric field $\vec{E}$ at a point on an equipotential surface must be perpendicular to the surface at that point ?

(i)

-  1

-  1

(a)

Note: Award full credit (2 marks) if any student draws both in the same diagram

(ii) If the field were not normal to the equipotential surface, it would have non-zero component along the surface. To move a unit test charge against the direction of the component of the field, work, would have to be done. But this is in contradiction to the definition of an equipotential surface. 1

<u>**Alternatively**</u>:

$$\Delta V = -\vec{E}.\vec{\Delta l}$$

$$0 = -E\,\Delta l \cos\theta$$

$$\theta = 90^0$$

1. In a region, the electric potential varies as $V = 10 - 50x$, where V is in volts and x in meters. The electric field in the region is : **1**

(A) 10 N/C along $+x$

(B) 10 N/C along $-x$

(C) 50 N/C along $+x$

(D) 50 N/C along $-x$

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1. In a region, the electric potential varies as V = 10 – 50x, where V is in volts and x in meters. The electric field in the region is : **1**

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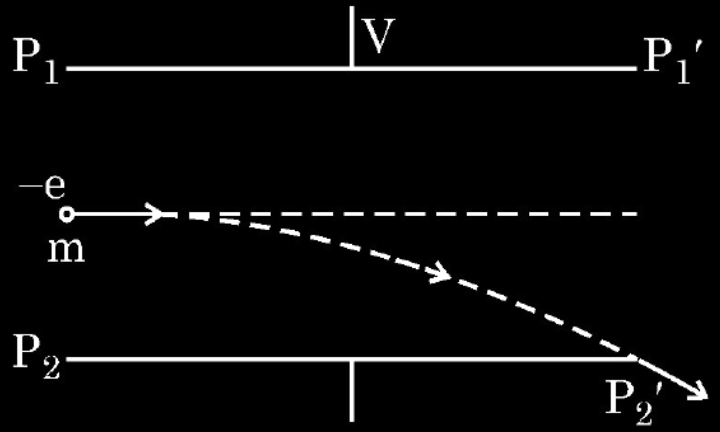
(C) 50 N/C along + x

24. Figure shows a narrow beam of electrons entering with a velocity of 3×10^7 m/s, symmetrically through the space between two parallel horizontal plates $P_1 P_1'$ and $P_2 P_2'$ kept 2 cm apart. CBSE 2026 **3**



If each plate is 3 cm long, calculate the potential difference V applied between the plates so that the beam just strikes the end P_2'.

Since $a = \frac{qE}{m} = \frac{qV}{mL}$	½
$t = \frac{x}{u_x}$	½
$y = \frac{1}{2}\left(\frac{qV}{mL}\right)\left(\frac{x}{u_x}\right)^2$	½
$V = \frac{2ymLu_x^{\ 2}}{ex^2}$	½
$V = \frac{2\times1\times10^{-2}\times9.1\times10^{-31}\times2\times10^{-2}\left(3\times10^{7}\right)^2}{1.6\times10^{-19}\times(3\times10^{-2})^2}$	½
$V = 2275\,\text{Volt}$	½

1. In a region electric field is given by $\vec{E} = 4\,x\hat{i}$ N/C. The potential difference between points A ($x = 1$ m) and B ($x = 3$ m), $(V_A - V_B)$ is **1**

(A) –16 V (B) 16 V

(C) – 8 V (D) 8 V

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(C) -8 V (D) 8 V

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$\boxed{\text{(B) 16V}}$

5. The electric potential for various points in x-y plane is given by $V = 1.0\ x^2 - 2.0\ y^2$, where V is in volts and x, y are in metres. The angle that the electric field at point (2.0 m, 1.0 m) makes with the positive x-axis is – **1**

(A) 45° (B) 90°

(C) 135° (D) 315°

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5. The electric potential for various points in x-y plane is given by $V = 1.0\,x^2 - 2.0\,y^2$, where V is in volts and x, y are in metres. The angle that the electric field at point (2.0 m, 1.0 m) makes with the positive x-axis is – **1**

(A) 45° (B) 90°

(C) 135° (D) 315°

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(C) 135^0